

Robust Cointegration Testing for Statistical Arbitrage

From estimator sensitivity to rolling stability and revalidation

Why does $\hat{\beta}$ fluctuate?

When is a cointegrating relation reliable?

1. Simple least-squares example: the coefficient can fluctuate although β is fixed

Assume the same regressors are observed repeatedly,

$$y_i = \beta x_i + \varepsilon_i, \quad E[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma_\varepsilon^2,$$

and only the realization of the noise changes. Then

$$\hat{\beta} - \beta = \frac{\sum_i x_i \varepsilon_i}{\sum_i x_i^2} \quad (\text{centered one-regressor case}).$$

Large Y-side shock

If $|\varepsilon_i|$ is large, the numerator can be large even though the underlying coefficient has not changed.

Little information in X

If $\sum_i x_i^2$ is small, the same disturbance produces a much larger change in the fitted coefficient.

First distinction: a moving estimate does not by itself imply a moving true coefficient.

2. Cointegration: coefficient fluctuation also changes the residual being tested

For a candidate cointegrating regression,

$$y_t = \alpha + \beta x_t + u_t, \quad x_t, y_t \sim I(1), \quad u_t \sim I(0),$$

the estimated coefficient defines the residual itself:

$$\hat{u}_t(\hat{\beta}) = y_t - \hat{\alpha} - \hat{\beta}x_t.$$

Coefficient side

Across windows we observe

$$\hat{\beta}_1, \hat{\beta}_2, \dots$$

which may fluctuate because of estimation sensitivity or because the underlying relation truly changes.

Residual side

Every new $\hat{\beta}_t$ produces a new residual series $\hat{u}(\hat{\beta}_t)$; therefore the stationarity statistic / p-value can fluctuate together with the coefficient estimate.

3. Empirical illustration: rolling coefficient fluctuation

Insert the empirical $\hat{\beta}_t$ plot here. The goal is to show visually that adjacent rolling estimates can move substantially before deciding whether the movement is statistical sensitivity or structural change.

ROLLING $\hat{\beta}_t$ / $|\Delta\hat{\beta}_t|$ PLOT

4. Empirical illustration: residual-stationarity evidence fluctuates too

Insert the residual-stationarity statistic or p-value here. This should use the same rolling windows as the coefficient plot so the two fluctuations can be compared directly.

ROLLING RESIDUAL TEST STATISTIC / P-VALUE PLOT

5. Mathematical formalization I: linking the supervisor's terminology to the model

Let $\hat{\beta}$ denote the exact OLS solution and $\tilde{\beta}$ the value returned numerically.

Intrinsic to the data / statistical estimation

Supervisor's wording

"data itself"

"limited sample"

low variation /
collinearity

Mathematical object

disturbance scale σ_ε^2 and realized ε

limited total information in $X'X$;
roughly $X'X \approx n\Sigma_X$

small eigenvalue of $X'X$: weak information in a coefficient direction

These are **not separate random mechanisms**: sample size and regressor geometry jointly determine how much information $X'X$ contains.

Intrinsic to the numerical method

"numerical error / ill-conditioning" refers to

$$e^{\text{num}} = \tilde{\beta} - \hat{\beta}.$$

Poor conditioning can amplify floating-point / algorithmic perturbations. This is distinct from the statistical variance of exact OLS.

Genuine relation change is separate again

$$\beta_{t+1}^* \neq \beta_t^*.$$

Even with abundant information and exact arithmetic, the fitted coefficient should move if the underlying relation itself changes.

6. Mathematical formalization II: fixed-window sampling uncertainty

Under local coefficient constancy and homoskedastic errors,

$$\text{Var}(\hat{\beta} \mid X) = \sigma_{\varepsilon}^2 (X'X)^{-1}.$$

For centered regressors, $n^{-1}X'X$ is the empirical second-moment / covariance scale, so approximately

$$\text{Var}(\hat{\beta} \mid X) \approx \frac{\sigma_{\varepsilon}^2}{n} \Sigma_X^{-1}.$$

Let q_k be an eigenvector of $X'X$ with eigenvalue λ_k . Then

$$\text{Var}(q_k' \hat{\beta} \mid X) = \frac{\sigma_{\varepsilon}^2}{\lambda_k}.$$

Under a Gaussian reference model,

$$P(|q_k'(\hat{\beta} - \beta)| > c \mid X) = 2\Phi\left(-\frac{c\sqrt{\lambda_k}}{\sigma_{\varepsilon}}\right).$$

$\sigma_{\varepsilon} \uparrow$
larger possible Y -side disturbances

$\lambda_k \downarrow$
less information in direction q_k

Together these are the SE: high noise + low information $\Rightarrow$ larger probability of a large sampling-induced coefficient deviation, even when the true β is unchanged.

7. Mathematical formalization III: the additional effect of moving the window

The fixed-window covariance asks how $\hat{\beta}$ varies if the same X is kept and the Y -noise is regenerated. Our actual rolling object is different:

$$\Delta\hat{\beta}_t = \hat{\beta}_{t+1} - \hat{\beta}_t,$$

because one case $(\mathbf{x}_-, y_-)$ leaves and one case $(\mathbf{x}_+, y_+)$ enters.

Let $G_t = X_t' X_t$, $r_- = y_- - \mathbf{x}_-'\hat{\beta}_t$, and $h_- = \mathbf{x}_-'\mathbf{x}_- G_t^{-1} \mathbf{x}_-$. The exact delete-add update is

$$\Delta\hat{\beta}_t = -\frac{G_t^{-1} \mathbf{x}_-}{1 - h_-} r_- + \frac{G_{t,-}^{-1} \mathbf{x}_+}{1 + h_+^{(-)}} r_+^{(-)}$$

with $G_{t,-} = G_t - \mathbf{x}_- \mathbf{x}_-'$ and $h_+^{(-)} = \mathbf{x}_+' G_{t,-}^{-1} \mathbf{x}_+$.

For intuition, along a weak-information direction $\mathbf{q}_k$,

$$q_k' \Delta\hat{\beta}_t \approx \frac{(q_k' \mathbf{x}_+) r_+ - (q_k' \mathbf{x}_-) r_-}{\lambda_k}.$$

Large rolling movement: little information in the old window (λ_k small) + entering/leaving $\mathbf{x}$ aligned with that direction + a large prediction residual.

8. Proposed qualification step: test the fitted relation forward, not on the same window

Detection / calibration window

1. Choose a pre-specified minimum window length $L_{\min}$.
2. Estimate the candidate relation $\hat{\beta}_0$.
3. Quantify its statistical sensitivity from $\hat{\sigma}_\varepsilon^2(X'X)^{-1}$ and the implied rolling-update scale.

Do **not** search the same sample for the window giving the smallest SE and then claim stability from that same sample; that introduces selection.

Forward proof-collection period

Freeze the candidate $\hat{\beta}_0$ and collect future observations. Require simultaneously:

1. **coefficient reliability:** future rolling movements are compatible with the predicted sampling / replacement sensitivity;
2. **residual validity:** $u_s^{(0)} = y_s - x_s'\hat{\beta}_0$ continues to behave as a stationary residual.

The two conditions can be evaluated over the **same future period**, but they answer different questions and should remain distinguishable.

9. From sensitivity qualification to structural-change monitoring

Reference / local-null layer

Assume temporarily

$$H_{0,t} : \beta_s^* = \beta_t^*$$

over the monitoring horizon. Use the current noise and information to obtain the expected distribution / scale of

$$\Delta \hat{\beta}_t.$$

This tells us how much movement is plausible *without* structural change.

Structural layer

If realized movements persistently exceed that reference behavior, investigate

$$\beta_{t+1}^* \neq \beta_t^*.$$

Classical CUSUM / sequential change-point methods provide the natural comparison literature.

Interpretation: first quantify estimator sensitivity; only then interpret excess movement as evidence for a changing relation. After an alarm: suspend, re-estimate, and revalidate the newly defined residual.

References: Brown, Durbin & Evans (1975); Horváth, Hušková, Kokoszka & Steinebach (2004); Wagner & Wied (2017); Trapani & Whitehouse (2020).

10. Results

Main empirical result / diagnostic plot

RESULT FIGURE

11. Research question and references

Can we quantify the rolling coefficient movement that is plausible from statistical sensitivity alone, and use excess movement together with residual stationarity to qualify and monitor a local cointegrating relation?

Core sequence

1. Explain fixed-window sampling uncertainty through noise and information.
2. Extend it to the actual delete–add rolling update $\Delta\hat{\beta}_t$.
3. Qualify the candidate forward using coefficient reliability and residual stationarity.
4. Monitor for persistent excess movement / breakdown; update and revalidate when needed.

Selected references

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